

AdaBoost: From First Principles

AdaBoost is an ensemble algorithm that builds many weak learners one after another. Each new learner pays more attention to the examples that previous learners got wrong.

Why AdaBoost Is Needed

A single simple model may be weak. AdaBoost shows that many weak models can combine into a strong model when each one focuses on previous mistakes.

Weak Learners

AdaBoost often uses decision stumps, which are trees with only one split. Each stump is weak alone, but many stumps can become powerful together.

Core Mechanism

AdaBoost starts with equal weights for all training examples. After each weak learner, wrongly classified examples receive higher weights, so the next learner focuses more on them.

Model Weight

A learner that performs well receives a higher vote weight. A learner that performs poorly receives a lower vote weight.

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - \text{error}}{\text{error}} \right)$$

Final Prediction

The final prediction is a weighted vote of all weak learners. Better learners have stronger votes.

Limitations

AdaBoost can be sensitive to noisy data and outliers because it keeps increasing attention on difficult examples.